

Data Structures - Class Test 2 - Answer scripts (typed)

Source: Mohler ASAG dataset (2011), CC-BY-4.0. Real student answers.

Question 6 (5 marks): How are infix expressions evaluated by computers?

Model answer: First, they are converted into postfix form, followed by an evaluation of the postfix expression.

Student 01

They are converted to post fix expressions. It still follow the order of precedence for the operators.

Student 02

They are first converted to postfix expressions and then evaluated.

Student 03

Computers cannot evaluate infix expressions, it must first convert infix to postfix expression then it can evaluate.

Student 04

By using parentheses to have order in the expression.

Student 05

In postfix notation.

Student 06

computers usually convert infix expressions to post fix expression and evaluate them using a stack.

Student 07

they are converted to postfix

Student 08

Infix expressions are interpreted as postfix expressions by implementing stacks.

Student 09

They are first converted into an equivalent postfix expression so it can be read by the computer.

Student 10

The computer translates the infix expression to a postfix expression, then solves.

Student 11

in postfix format

Student 12

they use parenthesis and a stack to determine which evaluations need to be made first

Student 13

By throwing variables into a postfix expression and operands onto a parentheses regulated stack until end parentheses and enough variables to operate on are encountered.

Student 14

by being passed to the function by a leading term

Student 15

i have an hard time explaining this so i will show how infix is evaluated instead. Start with an infix expression,

like, $(((5 \text{ plus } 2) \text{ multiplied by } 5) \text{ plus } (400 \text{ divided by } (2 \text{ plus } 3)))$, and push items until you get a ")" and once that happens, perform the operations until you reach an "(" ... with that complete, you will now have $((7 \text{ multiplied by } 5) \text{ plus } (400 \text{ divided by } (2 \text{ plus } 3)))$ as now the expression that will be evaluated ... perform last step again ... $(35 \text{ plus } (400 \text{ divided by } (2 \text{ plus } 3)))$ is now the stack ... repeat again ... $(35 \text{ plus } (400 \text{ divided by } 5))$ is now the stack after that ... repeat ... $(35 \text{ plus } 80)$ is now the stack, repeat again ... 115 is now the stack, and is returned.

Student 16

The infix expression is converted to postfix form

Student 17

by converting them to postfix expressions and putting the operations in a stack

Student 18

The computer converts the infix expression to postfix form. Then evaluates the postfix expression.

Student 19

infix expressions are converted to postfix expressions before they can be evaluated by a computer. Operands stay in the same order. Operators always follow the operands.

Student 20

infix expressions are pointers evaluated by reference to the memory location